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Abstractive Text Summarization Using BART

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***Abstract*—In the last recent years, there's a huge amount of data available on the internet, and is generated very rapidly. It is very difficult for human beings to analyze and extract useful information from huge data especially when the text is large in size and longer documents which increases the time to process and analyze the data, further it also increases the time taken to summarize it. To address this issue automatic text summarization is used. Text summarization is defined as creating a short, accurate, and fluent summary of a longer document. It summarizes the larger text without any human intervention. This paper will provide a mechanism where it does the text summarization quickly and effectively even for large data. We can use this model to summarize and extract important information from a large document or text based on our input. Here a concept of the Deep Learning model is used for text summarization which is called BART (Bidirectional and Auto-Regressive Transformer). it consists of both an encoder and decoder. The encoder and decoder are merged to form the BART algorithm. BERT is trained with a huge amount of unlabeled data to achieve the state of art results. The use of an attention mechanism in each layer of BERT makes the model much more popular as it highlights the important features of the input data. BERT performs the masked language modeling with the help of its several bidirectional transformer layers and predicts the missing values. On the other hand, decoder is used to predict the next token in a sentence. Merging both the encoder and decoder will form the BART model. Here we will compare the BART model with BERT, T5, and Roberta.**

Keywords— Text Summarization, BERT, BART, Roberta, Attention Mechanism, T5.

I. Introduction

Text Summarization alludes to the style of shortening long bits of text. The fundamental goal is to create an intelligible and familiar outline. There are two fundamental kinds of how to sum up the text in NLP: extraction-based synopsis. The extractive text rundown procedure includes catching the significant elements from the given text and consolidating them to make a synopsis. The extraction of expressions is made by the characterized norm without rolling out any improvements to the first text. This reflection additionally incorporates summarizing and shortening the text of the source record. At the point when reflection is applied for text synopsis in profound learning issues, it can beat the punctuation abnormalities of the extractive technique. The abstractive message rundown calculations produce new expressions and sentences that bear the most helpful data from the first reading material very much like people do. The objective of programmed message synopsis is to introduce the source message into a more limited rendition with semantics. The essential and significant benefit of utilizing an outline is to lessen the understanding time. Programmed text outline styles are significantly requested to address the always developing measure of text information accessible online to both better assist with finding appropriate data and consume related data rapidly.

Here, either the raw text or a text file is taken as input. After the text is extracted from the text file, the text undergoes various text preprocessing techniques like expanding contractions, converting the text into one case, stopword removal, removing digits and punctuations from the text, etc. After the preprocessing is completed, the preprocessed text is sent to the encoder. The duty of the encoder is to capture the contextually important features from the text using various attention mechanisms. Once the important features are extracted from the encoder, the text is sent to the decoder to find the next word probabilities. The duty of the decoder is to determine the output sequence based on the extracted contextual information from the encoder. The text obtained from the decoder is more readable than the text extracted from the encoder. The text obtained from the encoder may or may not follow the sequential order. But the decoder arranges the text in a sequential order.

II. Literature Survey

This paper proposes an abstractive multi-report outline framework called HMSumm [1] which is a blend of both extractive and abstractive synopsis draws near. To start with, it produces an extractive outline from different information archives, and that extractive synopsis is utilized to create the abstractive rundown. Then, at that point, determinantal point process (DPP) is utilized to manage overt repetitiveness which is a significant issue related with multi-record synopsis [2]. It removes the significant data from the given information reports with the goal that it have some control over the length of the information sentence and lessens the computational expense. A profound submodular network (DSN) is utilized to decide the extractive outline of the information reports. The achieved extractive synopsis is taken care of into BART and T5 pre-prepared models to create two abstractive outlines [3]. To appraise the exhibition of HMSumm, both human assessments and ROUGE-based appraisals are utilized and contrasted it and a few cutting edge styles. DUC 2002, DUC 2004, Multi-News, and CNN/Daily Mail datasets are utilized to assess the calculations. The exploratory outcomes show that HMSumm beats the subsidiary cutting edge calculations [4].

The primary objective of this examination is to furnish a general comprehension and knowledge of the components of ongoing brain networks based abstractive text rundown models with an exceptional survey [5], as well as to give a consciousness of the difficulties and issues with these frameworks. In this paper, different brain networks based abstractive outline models have been introduced. The proposed calculated structure incorporates five key components recognized as encoder-decoder design, systems, preparing techniques and advancement calculations, dataset, and assessment measurements. Models utilizing a transformer-based encoder-decoder plans have been viewed as the new cutting edge. In light of the information gained from the review, this article proposes the utilization of pre-prepared language models like BART and MASS in

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supplement with brain network engineering for abstractive synopsis assignments [6].

III. METHODOLOGY

Text summarization is defined as generation of short, accurate and fluent summary of the given longer text. There are two types of text summarization

- Extractive text summarization
- Abstractive text summarization

A. Extractive text summarization

It is the basic text summarization technique which was developed first. The main objective of this technique is to capture the important sentences of the text and it will be added to the final summary. Here the summary obtained contains the same sentences from the original text.

B. Abstractive text summarization

This type of text summarization is an extension to the extractive text summarization. This will also capture the important sentences from the longer text and redefines it and produce in a new way. This type of summarization also makes sure that summary generated as short as possible.

C. Expanding contractions

Contraction in text preprocessing refers to shorter representation of words. The contraction needs to expanded to make perform better analysis on text.

D. Lower casing

If all the text is in the same case, then it becomes easy for the machine to interpret the text. This is because the lower-case text and upper-case text are treated differently by the machine. So, it's better to convert the text either lower case or upper case before performing operations on text

E. Removing punctuations

This is also one of the widely used pre-processing technique. The objective of this technique is to remove punctuations from the text. Though the punctuations had some weight with reference to the context of the text but it has very lesser weights when performing the analysis of the textual data. So, we need to remove punctuations from the text before performing any operations on the text. In total there are 32 punctuations in English.

Example: - #, @, %, ……...

F. Remove words and digits containing digits

Sometimes there is a probability that words and digits combined and represented in the form of text. This will create a problem for machines as well as humans to understand and interpret the text. hence, our task is to remove the words which are combined with digits. Generally, this type of words is difficult to process. so it's better to remove them or replace them with an empty string.

Example: - Te123xt, Lan73guage, etc.

G. Removal of stopwords

Stopwords are defined as the words that occur most frequently which does not provide any valuable information with reference to the context of the text. So, we need to remove the stopwords before performing any operations on the text

Example: - They, there, this, where.

IV. BERT

BERT stands for Bidirectional encoder representations from transformers. It is a pre-trained model where we can easily perform the NLP tasks. Bert is introduced to overcome the drawbacks of RNN and LSTM's.

A. Disadvantages of RNN

- In RNN training consumes lot of time and it is a difficult task
- More chance of occurrence of vanishing gradient and exploding gradient problems
- It does not effectively handle the long sequences

B. Disadvantages of LSTM

- Still LSTM cannot reduce gradient problems completely
- LSTM are very likely to overfitting
- It requires more resources like it will take longer time and it also require lot of memory.

C. Advantages of BERT

- It is a pre trained model on huge data and it no longer require further training
- It has a power encoder and decoder embedded with transformer layers
- It is better than RNN and LSTM's
- The summary id generated based on the important sentences from the given longer text
- It performs the operations parallelly, where as in RNN and LSTM each word is processed one after another.

BERT is trained with two different strategies which are masked language modelling and next sequence prediction. In Masked Language Modelling a random group of tokens from the input text is selected and replaced with a special token which is a <mask>. The objective of BERT is to predict the masked tokens [7]. usually, BERT selects 15 % of the total text for replacement. in the 15% of the selected text, 80 % of the tokens are masked and 10% of tokens are left unchanged and the remaining 10% of tokens are replaced by a random vocabulary that suits the original text. The objective of Next Sentence Prediction is to judge whether two sentences follow each other or not. in the original text. Here the input is given in the form of sentences where each sentence is preceded by CLS tag and succeeded by SEP tag. The entire text is processed based on the sentences.

D. Token embeddings

After the input text is divided into sentences now, we need to find the embeddings of each token in the sentence. This is because the machine cannot understand the textual representation directly. So, we need to convert them to numerical representation to make it understandable by the machine. Here each token has a unique embedding and corresponding token embeddings are passed to the next stage.

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E. Segment embeddings

After the token embeddings now it's time to calculate the segment embeddings. The objective of segment embedding is to make the machine know to which sentence the token belongs to. This is just to make sure that which tokens belongs to which sentences. If the token belongs to sentence 1, then make it 0. If the token belongs to sentence 2 then make it 1

F. Positional encodings

Though the embedding is generated from the previous steps but they do not consider the position of the words. To make it embeddings clearer we add positional encoding. With the positional encoding we can able to distinguish the sentences that has same tokens.

For example, let us take two sentences

Sentence-1: - He has a black Ferrari. He also has a white cat.

Sentence-2: - He has a black & white Ferrari

If we calculate embeddings from the input embeddings, we get similar type of embeddings for the two sentences. But in reality, the two statements are different to each other. We can calculate the positional embeddings using sin and cos functions.

"P (K, J)" is the positional function that is used to map the tokens in the input sequence "n" is the user-defined value that can be set by the user. usually, it is set to 10000. "i" represents the current token

All the token embeddings, segment embeddings, positional embeddings are added element wise to produce a single representation. The obtained single representation is passed to the encoder for the extractive summarization.

G. Encoder

The aim of an Encoder is to take the entire text as an input and to identify the tokens which are important. In short, the role of encoder is to capture the contextual information from the given text. This identification of contextual information from the text is done by using various attention layers in the encoder. The embeddings are sent to the multi head attention layer where it captures the important information from the text.

H. Multi-head attention

Generally, Attention is basically a mechanism which provides important tokens in the input. A Multi-Head Attention Layer can be considered a group of parallel Attention Layers.

For example, take the sentence 'He sat on the chair & it broke' Attention 1: - It' with chair Attention 2: - It with 'He'

In this way various statements require multiple attentions to capture the better-quality contextual information. So multi head attention makes our task easier by having multiple attentions in the sentence.

I. Add & normalize

The duty of this layer is to normalize the output vector. When we concatenate all the embeddings of token, we end up having the vector which is of more dimensions. So, we need to normalize it. Add & normalize layer normalizes the output vector obtained from the multi-head attention layer. Here the obtained vector is multiplied with another vector to get the same dimensions. Finally the embeddings are sent to fully connected layer given text and the decoder is used to find the next word probabilities so that the text can be represented in human understandable and readable form. In BART multi head attention is used in encoder and masked multi head attention is used in decoder [8]. The aim of using a Decoder is to determine the output sequence of sentences based on the extracted conceptual information from the encoder

J. Masked multi head attention

After the conceptual information is extracted from the encoder, now it's time to generate the abstractive summary. The conceptual sentences extracted may contain noise and they are not in a sequential order. So, decoder arranges the extracted conceptual information in a sequence so that it adds more meaning to the summary. The decoder takes the input text and output from the decoder and predicts the next word probabilities. The decoder is trained in such a way that it is able to predict the next word of the sequence given the input text and attention obtained from the encoder. Now the problems arise at the first token as there are no previous tokens [9-11]. So, the BOS (beginning of the statement) is added at he begging which specifies the start of the statement as well as the it is used to predict the first token given that BOS is the previous token. In masked multi- head attention layer, attention is applied on tokens up-to current position and not applied on future tokens.

For example If we want to generate the embedding of third vector which is y3, then we consider the input embeddings up-to x2 and the original text to generate the embeddings of y3. Here we will not consider y4 and y5 to generate the embeddings. In self attention we consider all the input embeddings but here we apply mask to certain tokens. That's why it is named as masked self-attention.

V. BART

A. Decoder

Bart consists of both encoder and decoder. The encoder is responsible for fetching the important information from the given text and the decoder is used to find the next word probabilities so that the text can be represented in human understandable and readable form. In BART multi head attention is used in encoder and masked multi head attention is used in decoder. The aim of using a Decoder is to determine the output sequence of sentences based on the extracted conceptual information from the encoder

B. Masked multi head attention

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C. T5 self attention mechanism

Both the T5 and BART contains both encoder and decoder. But in T5 self-attention mechanism is used but where as in BART multi head attention is used. Generally, word embeddings are vectors that represent the semantic meaning of a word. In general, two similar Words may have similar embeddings. But in reality, the meaning of the words depends on the further a words and sentences. For example, let us consider "BANK of the river"[13-14]. Here the word bank has correlation with river as well as financial transactions. we humans can understand it depending on the next tokens in the sentence. but machine can't understand it properly. So, this is the reason why we need to used self-attention in order to distinguish the meanings of words with respect to next tokens.

VI. ROBERTRA

ROBERTA is abbreviated as Robustly Optimized BERT Pre- training Approach. it was developed by Facebook. ROBERTA deals with the optimization of BERT and to make sure that it will consume lesser time for pre-training. The following are the improvements made in ROBERTA when compared to BERT.

- Removal of next sentence prediction
- Training with larger batch sizes and long sentences
- introduced the dynamic mask strategy

A. Removal of next sequence prediction

In next sentence prediction, the model is trained in such a way that it needs to predict whether the two sentences follow each other or not. The developers had done various experiments with the addition and removal of next sequence prediction in various NLP tasks. Though it shows slightly lesser performance for text summarization tasks it shows improved performance with other NLP tasks and it is a time-consuming process so the NSP is removed in ROBERTA.

B. Training with larger batch sizes and long sentences

Roberta was trained with bigger batch sizes and longer sentences when compared to BERT resulting in achieving a higher performance when compared to BERT. it will increase the purpose of masked language modelling to a greater extent and also accuracy as well.

C. Introduced dynamic mask strategy

In BERT static masking is used when it masks some of the tokens of the input text during text pre-processing and also masking is done usually once resulting in a singe static masking. To avoid the single static masking dynamic masking is introduced where the training data is replicated and masked in ten different ways using different masking strategies. Here the model will perform better in predicting the masked tokens when compared to BERT.

VII. RESULTS AND DISCUSSION

A. Rouge score

ROUGE is abbreviated as Recall-Oriented Understudy for Gisting Evaluation. It is the most widely used evaluation metric for text summarization as well as other machine translations. Here the score is produced by comparing the original text and machine-generated summary. ROUGE uses the concept of n-gram to generate the score for text summarization. If it uses the unigrams then it is said to be ROUGE-1. If it uses the bigrams then it is said to be ROUGE-

If it uses the trigrams then it is said to be ROUGE-3. If it uses the longest common subsequence then it is said to be ROUGE-L. We can calculate recall, precision and f1 score for each score

B. Rouge-1

Recall= no of unigram matches / no of words in the reference text

Precision= no of unigram matches / no of words in the generated summary

F1=2*recall*precision / recall + precision

C. Rouge-2

Recall = no of bigram matches / no of words in the reference text

Precision = no of bigram matches / no of words in the generated summary

F1 =2*recall*precision / recall + precision

D. Rouge-3

Recall= no of trigram matches / no of words in the reference text

Precision= no of trigram matches / no of words in the generated summary

F1=2*recall*precision / recall + precision

E. Rouge-L

Recall= no of words in longest common subsequence / no of words in the reference text

Precision= no of words in longest common subsequence / no of words in the generated summary

F1=2*recall*precision / recall + precision

For example, consider the text,

Reference: I am at the office

Generated summary: am am am am am if we calculate the precision we get precision: 1+1+1+1+1 / 5 =1

Here we get precision as 1 which says that generated and actual text is similar but in reality, both the texts are different and had different meanings

F. Bleu Score

In rouge score there is problem with the precision, so to overcome this disadvantage a new score was introduces which is called as BLEU score. Here the precision is modified in such a way that it will overcome the drawback of rouge score. The formulae for modified precision is given as,

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Modified precision = no of word matches / total no of words in summary.

For the above example if we calculate the precision according to BLEU score, we get

Modified precision = 1 / 5 = 0.2

Here it will not consider the duplicates, and "am" is present only once in the original text. So, word matches are given as 1.

G. *problems in modified precision*

Still there are some problems in modified precision. let us take the following original and generated text.

Original text: I am at the office

Generated text: I am

Modified precision: 1+1 / 2 =1

Here we get modified precision as 1 which specifies both generated and original text are same, but in reality, both the texts are different and had different meanings. To overcome this limitation in modified precision, a penalty is imposed which is called brevity penalty.

The experiments were carried out on four different models, which are: BERT, BART, ROBERTA, and T5. Out of the four models, BART performs well when compared to the other three models. This is due to the fact that the BARTcontains both an encoder and a decoder. The main reason behind the better summarization of BART is multi-headattention. Multi-head attention is a group of self-attention where it can pay attention to different parts of a sentence. Because of this, the model learns the text better when compared to the other three models. Fig. 1. Shows the text summarization model performance Graph.

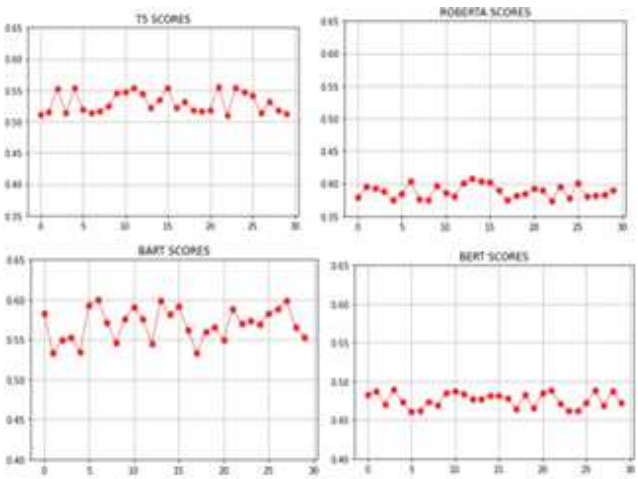

Fig. 1. Text Summarization Model Performance Graph

Though ROBERTA is an optimized version of BERT and trained with longer sentence text when compared with BERT, the model did notperform well in summarization tasks due to the removal of next sequence prediction. The BERT model is trained with both masked language modelling and next sequence prediction. In ROBERTA, the next sequence prediction is removed due to which it can't perform summarization tasks efficiently.

TABLE. I. PERFORMANCE OF ROGUE-1, ROUGE-2,ROUGE-3

Model	Rouge-1	Rouge-2	Bleu
Bert	40	31	0.48
Roberta	37	23	0.38
T5	36	25	0.54
Bart	58	46	0.56

T5 model achieves slightly lower score when compared with BART because here self-attention is used. because of this, it will handle the text with same context effectively. Though it has both an encoder and a decoder, due to self-attention, it can't focus on all parts of the sentence. It can only focus on a few parts of the sentence. Fig 2 depicts comparison of different models.

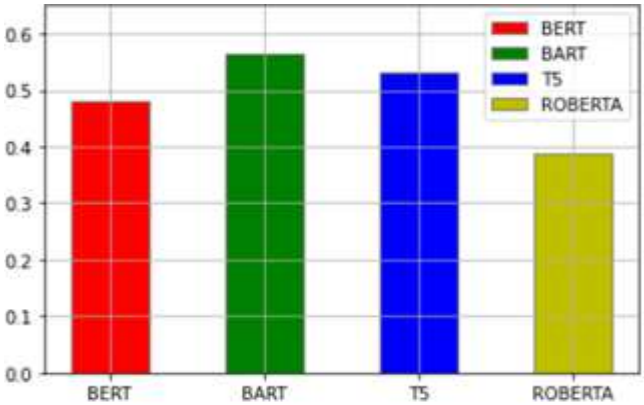

Fig. 2. Comparison of different models

VIII. CONCLUSION

Due to the exponential growth of web textual data due to the Internet's quick expansion, tasks like paper records, text classification, and information retrieval have been significantly hampered. Summarizing text automatically is a very essential way to address this issue. The main function of ATS is to automatically create a brief and understandable summary by extracting the main ideas from the source text. Recently, deep learning-based abstractive summarization models have been created in an effort to better balance and enhance these two characteristics. This study will present a system that can quickly and accurately summarize text, even for massive amounts of data. Using our input, utilize this method to summarize and extract crucial information from a lengthy document or text. Furthermore, a Deep Learning model approach known as BART is utilized for text summarization. The BART algorithm is created by combining the encoder and decoder. Additionally, the evaluation of the BART model using BERT, T5, and Roberta is shown in the result analysis. In future, this research can be extended by analyzing various learning models to improve the effectiveness.

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